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United States Patent Application Publication

20250283136

Kind Code

A1

Publication Date

September 11, 2025

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SPATIALLY ADDRESSABLE CONTROL OF POLYMERASE ACTIVITY

Abstract

Multiple polynucleotides having different, arbitrary sequences are synthesized on the surface of an array by spatial control of polymerase activity. The polymerase is a template-independent polymerase such as terminal deoxynucleotidyl transferase (TdT). Spatial control of polymerase activity is implemented by localized changes in redox-pH conditions. A single species of nucleotide is added and incorporated on growing polynucleotide strands at locations on the array where the polymerase is active. A washing step removes the polymerase and free nucleotides. This process may be repeated multiple times changing both the location of polymerase activity and the species of nucleotide thereby synthesizing different polynucleotides in parallel on the surface of the array. Polymerase activity may be regulated by removing a blocking group attached to a His-tag sequence on the polymerase, a change in pH, or release of encapsulated inhibitors.

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Family ID: 77821980

Appl. No.: 19/215304

Filed: May 21, 2025

Related U.S. Application Data

parent US division 17086055 20201030 parent-grant-document US 12344876 child US 19215304

Publication Classification

Int. Cl.: C12P19/34 (20060101); C12N9/12 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional of U.S. patent application Ser. No. 17/086,055, filed Oct. 30, 2020, the content of which is hereby expressly incorporated herein by reference in its entirety.

BACKGROUND

[0002] Synthetic oligonucleotides, also referred to as polynucleotides, such as deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) have uses in medicine, molecular biology, nanotechnology, data storage, and other applications. Enzymatic polynucleotide synthesis has emerged as an alternative to the long-standing nucleoside phosphoramidite method for synthesis of polynucleotides. Enzymatic polynucleotide synthesis is performed with a template-independent polymerase such as terminal deoxynucleotide transferase (TdT) rather than a series of chemical reactions. Enzymatic polynucleotide synthesis has advantages over the nucleoside phosphoramidite method because it is performed in an aqueous environment and does not use toxic organic chemicals. Enzymatic polynucleotide synthesis also has the potential to create longer polynucleotides than the nucleoside phosphoramidite method.

[0003] However, template-independent polymerases add nucleotides in an unregulated manner. These polymerases can add any available nucleotide and can create random sequences if provided with multiple types of nucleotides. If only a single species of nucleotide is present, the nucleotide may be added repeatedly creating variable length homopolymers. Thus, it is challenging to precisely control the base-by-base sequence of polynucleotides created through enzymatic polynucleotide synthesis. In contrast, with the established nucleoside phosphoramidite method, each synthesis cycle reliably adds only a single, specific nucleotide.

[0004] Techniques for highly parallel and automated enzymatic-based methods are clearly desirable for many applications such as digital data storage. However, controlling the enzymes used for polynucleotide synthesis brings unique challenges that are not present in chemical phosphoramidite synthesis. This disclosure is made with respect to these and other considerations.

SUMMARY

[0005] This disclosure provides methods and devices for de novo array-based enzymatic solid-phase synthesis of polynucleotides. Spatially addressable control of polymerase activity on the surface of an array allows for parallel synthesis of multiple polynucleotides with different sequences. The spatial control is provided by regulating the activity of a template-independent polymerase such as TdT.

[0006] Spatial addressability, the ability to turn polymerization “on” or “off” at selected locations on an array, enables the synthesis of polynucleotides of different sequences on the same array. Polymerases are activated on a selected location on the surface of the array, a single species of nucleotide is provided, and that nucleotide is incorporated into growing polynucleotide strands at the selected location. Nucleotide addition does not occur to an appreciable degree at locations where the polymerase is not activated. A reaction reagent solution covering the array is washed away and polynucleotide extension stops. This process is repeated multiple times. The selected location and the selected nucleotide species may both be independently changed in subsequent rounds of addition. With this technique, polynucleotides with different, arbitrary sequences are synthesized on the surface of the array.

[0007] In a first implementation, activity of the template-independent polymerase is regulated by the presence or absence of a blocking group attached to the enzyme. The blocking group may be a protein or polymer that is attached to the template-independent polymerase by a His-tag. The

blocking group sterically hinders access to the active site on the polymerase preventing activity. His-tags are complexed to a ligand on the blocking group with divalent metal cations such as Cu^{2+} . Changing the oxidation state of the metal cation from $2+$ breaks the His-tag complex and releases the blocking group from the template-independent polymerase. Thus, the polymerase becomes active only in those locations where the oxidation state of the metal cation is changed. The oxidation state of the metal cation may be changed by redox reactions initiated through electrodes, addition of chemical redox reagents, or excitation of photoredox catalysts. Activation of the polymerase is confined to the selected location by an excess of divalent metal cations in the reaction reagent solution. The divalent metal cations cause the blocking group to reattach to the His-tag if a polymerase moves away from the selected location.

[0008] In a second implementation, activity of the template-independent polymerase is regulated by pH. The polymerase and selected nucleotide species may be provided in a solution with a pH that inactivates the polymerase—an “unsuitable pH.” The pH of the solution at a selected location on the surface of the array is changed from the unsuitable pH to an optimum pH for the polymerase. The polymerase becomes active at the selected location but not at other locations due to the localized pH change. The pH of the solution may be changed by activation of electrodes, addition of an acid or base, or photoactivation of a photoacid or a photobase. The change in pH is confined to the selected location by a buffer in a reaction reagent solution.

[0009] In a third implementation, activity of the template-independent polymerase is regulated by the presence of enzyme inhibitors such as adenosine triphosphate (ATP). The enzyme inhibitors are encapsulated or otherwise inactivated in solution. Thus, the polymerase is active at all locations where the enzyme inhibitors are encapsulated. Release of the enzyme inhibitors from encapsulation by a change in pH inactivates the polymerase. The enzyme inhibitors may be released from encapsulation at a selected location by a localized change in pH. The pH may be changed by any of the techniques mentioned above. In this implementation, unlike the first and second implementations, the polymerase is inactivated rather than activated at the selected location. This technique may lead to some level of nucleotide incorporation across the entire array but there will be much less at the locations where inhibitors are released. The different lengths of polynucleotide extension may be used to encode information even if they do not have precisely specified base-by-base sequences.

[0010] To prevent incorporation of nucleotides across the entire surface of the array before release of enzyme inhibitors, a different mechanism may initially keep the polymerase inactive. For example, the temperature of a solution containing the template-independent polymerase may be kept at a low temperature at which the polymerase has substantially reduced activity. After the enzyme inhibitors are released from encapsulation, the temperature is increased, and the polymerase becomes active. Nucleotides are added to the growing polynucleotide strands at locations other than the selected location. At the selected location, nucleotides are not added or added at a greatly reduced rate due to the presence of inhibitors.

[0011] Array-based synthesis of polynucleotides improves the scalability and throughput of previous enzymatic synthesis techniques that use beads in a test tube for solid-phase synthesis. All polynucleotides synthesized in the same test tube, plate well, or reaction chamber are exposed to the same conditions and thus will have the same sequence of nucleotides. This requires a physically separate reaction environment for each unique polynucleotide sequence that is synthesized. However, array-based synthesis techniques in which localized reaction environments (e.g., redox conditions or pH) can be changed on the surface of the array provide the ability to synthesize polynucleotides with different sequences on the same array. This design is more compact and requires less physical manipulation than a comparable system in which each unique polynucleotide sequence must be created in a different tube or well.

[0012] This disclosure also provides a device for de novo synthesis of polynucleotides using an array and a reaction reagent solution containing template-independent polymerase. This device may

include fluid delivery pathways for adding the reaction reagent solution and selected species of nucleotides to the surface of the array. The device may control redox-pH conditions on the surface of the array through any number of different techniques. In an implementation, the array is a microelectrode array with individually addressable electrodes that can change a local redox-pH environment upon activation. In an implementation, a targeted fluid deposition instrument such as a chemical inkjet printer may be used to add small volumes of redox reagent, acid, or base to specific locations on the surface of the array. In an implementation, a light source capable of inducing a photoredox reaction by exciting a photoredox catalyst or changing pH by activation of a photoacid or photobase may be directed onto specific locations on the surface of the array.

[0013] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter nor is it intended to be used to limit the scope of the claimed subject matter. The term “techniques,” for instance, may refer to system(s) and/or method(s) as permitted by the context described above and throughout the document.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The Detailed Description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items.

[0015] FIG. 1 shows solid-phase polynucleotide synthesis on an array using spatial control of polymerase activity.

[0016] FIG. 2 shows use of a blocking group attached by a His-tag sequence to regulate polymerase activity.

[0017] FIG. 3 shows an array for solid-phase polynucleotide synthesis on which spatial control of polymerase activity is regulated by pH.

[0018] FIG. 4 shows an array for solid-phase polynucleotide synthesis on which spatial control of polymerase activity is regulated by release of encapsulated enzyme inhibitors.

[0019] FIG. 5 is a flow diagram showing an illustrative process for solid-phase polynucleotide synthesis using redox-pH conditions to provide spatial control of polymerase activity.

[0020] FIG. 6 is an illustrative device for de novo synthesis of polynucleotides.

DETAILED DESCRIPTION

[0021] This disclosure provides techniques for solid-phase de novo synthesis of polynucleotides with arbitrary sequences by location-specific control of template-independent polymerase activity. Many existing techniques for enzymatic synthesis of polynucleotides control polymerization by regulating availability of nucleotides rather than polymerase activity. The techniques of this disclosure are thus able to use unmodified nucleotides rather than requiring nucleotides attached to a blocking group. This disclosure describes multiple techniques for controlling polymerase activity in a location-specific manner. These techniques include use of a His-tag to attach a blocking group to a polymerase, regulation of pH to prevent polymerase activity, and selective release of encapsulated enzyme inhibitors. Moreover, the techniques of this disclosure regulate enzyme activity without changing the availability of a metal cofactor necessary for functioning of the template-independent polymerase. Availability of a metal cofactor refers to the presence of a suitable metal cofactor in the appropriate oxidation state at sufficient concentration for polymerases activity. A metal cofactor may be made unavailable by multiple techniques including, but not limited to, sequestration such as with a ligand or by changing its oxidation state.

[0022] There are many uses for synthetic polynucleotides having specified sequences such as basic

research, medicine, and nanoengineering (e.g., DNA origami). One relatively recent application for synthetic polynucleotides is digital data storage. Polynucleotides such as DNA may be used to store digital information by designing a sequence of nucleotide bases that encodes the zeros and ones of the digital information. There are various techniques and encoding schemes known to those of skill in the art for using nucleotide bases to represent digital information. See Lee Organick et al., *Random Access in Large-Scale DNA Data Storage*, 36:3 Nat. Biotech. 243 (2018) and Melpomeni Dimpoulou et al., *Storing Digital Data Into DNA: A Comparative Study of Quaternary Code Construction*, ICASSP Barcelona, Spain (2020). Advantages of using polynucleotides rather than another storage media for storing digital information include information density and longevity. The sequence of nucleotide bases is designed on a computer and then polynucleotides with those sequences are synthesized. The polynucleotides may be stored and later read by a polynucleotide sequencer to retrieve the digital information.

[0023] Polynucleotides, also referred to as oligonucleotides, include both DNA, RNA, and hybrids containing mixtures of DNA and RNA. DNA includes nucleotides with one of the four natural bases cytosine (C), guanine (G), adenine (A), or thymine (T) as well as unnatural bases, noncanonical bases, and/or modified bases. RNA includes nucleotides with one of the four natural bases cytosine, guanine, adenine, or uracil (U) as well as unnatural bases, noncanonical bases, and/or modified bases. Nucleotides include both deoxyribonucleotides and ribonucleotides covalently linked to one or more phosphate groups.

[0024] Template-independent polymerases are DNA or RNA polymerases that perform de novo oligonucleotide synthesis without use of a template strand. Currently known template-independent polymerases include TdT and tRNA nucleotidyltransferase. TdT as used herein includes both the full-length wild-type enzyme, as well as modified enzymes that are truncated or internally modified. One example of modified TdT is provided in U.S. Pat. No. 10,059,929. An example of truncated TdT is provided in U.S. Pat. No. 7,494,797. Thus, template-independent polymerase as used herein includes full-length wild-type, truncated, or otherwise modified TdT, tRNA nucleotidyltransferase, and any subsequently discovered or engineered polymerases that can perform template-independent synthesis of polynucleotides. Template-independent polymerase as used herein does not encompass modifications that render an enzyme incapable of performing nucleotide polymerization.

[0025] TdT is a protein that evolved to rapidly catalyze the linkage of naturally occurring deoxynucleotide triphosphates (dNTPs). Native TdT is a very efficient enzyme. It has been demonstrated that TdT can polymerize extremely long homopolydeoxynucleotides of 1000 to 10,000 nucleotides in length (see Hoard et al., *J. of Biol. Chem.*, 1969 244(19):536373; F.J. Bollum, *The Enzymes*, Volume 10, New York: Academic Press; 1974. p. 141-71; Tjong et al., *Anal. Chem.* 2011, 83:5153-59. TdT adds nucleotides indiscriminately to the 3' hydroxyl group at the 3' end of single-stranded DNA. TdT performs unregulated synthesis adding any available dNTP. TdT uses an existing polynucleotide referred to as an "initiator" as the starting point for synthesis. Initiators as short as three nucleotides have been successfully used with TdT for enzymatic synthesis of DNA. Suitable initiator length ranges from three nucleotides to about 30 nucleotides or longer. Initiators may be single stranded or double stranded. Double stranded initiators may have a 3' overhang or they may be blunt ended or they may have a 3' recessed end. During polymerization, the template-independent polymerase holds a DNA strand (which initially is only the initiator but grows as synthesis proceeds) and adds dNTPs in a 5'-3' direction. TdT activity is maximized at approximately 37° C. and performs enzymatic reactions in an aqueous environment.

[0026] Because TdT performs unregulated synthesis, using this enzyme to create a polynucleotide with a pre-specified arbitrary sequence requires regulation and control of the TdT activity. One technique to regulate TdT activity is limiting the available nucleotides to only a single type of deoxynucleoside triphosphate (dNTP) or nucleoside triphosphate (NTP) (e.g., only deoxyadenosine triphosphate (dATP), deoxycytidine triphosphate (dCTP), deoxyguanosine triphosphate (dGTP),

deoxythymidine triphosphate (dTTP), adenosine triphosphate (ATP), cytidine triphosphate (CTP), guanosine triphosphate (GTP), or uridine triphosphate (UTP)). Thus, providing only one choice forces the polymerase to add that type of nucleotide.

[0027] However, this does not prevent TdT from adding the nucleotide multiple times thereby creating homopolymers. Techniques for limiting homopolymer creation by TdT include using nucleotides with removable protecting groups that prevent addition of more than one nucleotide at a time. Examples of techniques that use blocking groups attached to nucleotides are described in U.S. Pat. Nos. 10,059,929 and 10,683,536. These techniques require specially modified dNTPs.

[0028] Another technique for enzymatic synthesis uses TdT enzymes each tethered to a single dNTP by a cleavable linker. See Sebastian Palluk et al., *De novo DNA synthesis using polymerase-nucleotide conjugates*, 36(7) Nature Biotechnology 645 (2018) and WO 2017/223517 A1. In this system dNTPs are modified by attachment to TdT.

[0029] A third technique for nucleotide synthesis using TdT regulates polymerization by including the enzyme apyrase, which degrades nucleoside triphosphates into their TdT-inactive diphosphate and monophosphate precursors. In this technique, apyrase limits polymerization by competing with TdT for nucleoside triphosphates. See Henry H. Lee et al., *Terminator-free template-independent Enzymatic DNA Synthesis for Digital Information Storage*, 10:2383 Nat. Comm. (2019) and WO 2017/176541 A1. Although unmodified nucleotides may be used in this technique, it is nucleotide availability rather than enzyme activity that is controlled.

[0030] Although the above techniques may be used to limit the “extension length” or average number of nucleotides added during a cycle of synthesis, they do not describe synthesis of multiple polynucleotides with different sequences on a single array. These solid-phase enzymatic nucleotide synthesis techniques involve initiators attached to beads in a test tube or other discrete reaction chamber. The reaction chamber is flooded with an aqueous solution containing TdT and only one type of dNTP. Once coupling has taken place, the TdT and any free dNTPs are washed away. The beads are incubated in a second step with TdT and a different dNTP. The process continues creating DNA molecules with sequence specified by the order in which the different dNTPs are added. Depending on the control technique used, TdT may add a single nucleotide or an uncontrolled number of the same nucleotide during each cycle synthesis. This process does not scale well for applications that require high throughput synthesis of multiple polynucleotides with different sequences.

[0031] There are techniques that provide spatial control of template-independent polymerase on an array. These techniques do so by regulating the availability of metal cofactors that are necessary for enzyme activity. One technique keeps the metal cofactors in an inactive state by caging with DMNP-EDTA and releases the metal cofactors at specific locations by exposure to patterned UV light. Diffusion of the metal cofactors is controlled by providing an excess of the caging molecules. The TdT and nucleotides are provided in a standard synthesis master mix. See Howon Lee et al., *Photon-directed Multiplexed Enzymatic DNA Synthesis for Molecular Digital Data Storage*, bioRxiv 2020.02.19.956888.

[0032] A different technique also by the inventors of this application, controls the oxidation state of metal enzyme cofactors. The metal cofactors are changed from an oxidation state of +2 that complexes with the enzyme to a different oxidation state that does not. Template-independent polymerase is inactive unless metal cofactors with an oxidation state of +2 are available. Spatial control of the oxidation state is achieved by activation of electrodes on a microelectrode array, controlled addition of redox reagents, or other techniques. Diffusion of the metal cofactors in the +2 oxidation state is controlled by scavenger molecules that either change the oxidation state or sequester the metal cofactors. See U.S. Pat. Pub. No. US2021/0058216 filed on Aug. 16, 2019, with the title “Regulation of Polymerase Using Cofactor Oxidation States.” However, control of metal cofactor availability is only one way of regulating polymerase activity.

[0033] Detail of procedures and techniques not explicitly described or other processes disclosed in

this application are understood to be performed using conventional molecular biology techniques and knowledge readily available to one of ordinary skill in the art. Specific procedures and techniques may be found in reference manuals such as, for example, Michael R. Green & Joseph Sambrook, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Laboratory Press, 4^{sup}.th ed. (2012).

[0034] FIG. 1 shows an illustrative representation of solid-phase synthesis on an array **100** in which the location of nucleotide addition is regulated by controlling activity of a template-independent polymerase. The array **100** provides a solid support for solid-phase synthesis of polynucleotides. Solid-phase synthesis is a method in which molecules are covalently bound on a solid support material and synthesized step-by-step in a single reaction vessel. The polymerase is a template-independent polymerase such as TdT. The template-independent polymerase may be obtained from a number of sources such as isolation from calf thymus or a recombinant source (e.g., a genetically modified *E. coli* strain). In some implementations, the template-independent polymerase is synthesized chemically or using recombinant techniques in order to add a His-tag.

[0035] The array **100** may be made of any material that is capable of anchoring polynucleotides. The array **100** may be formed from a silicon chip, glass (e.g., controlled porous glass (CPG)), an insoluble polymer, or other material. The array **100** being a generally flat two-dimensional surface provides for addressable, site-specific manipulations at specified locations (e.g., represented in terms of x- and y-coordinates) on the surface of the array **100**. The array **100** may be an electrochemically inert surface or it may include an array of individually addressable microelectrodes.

[0036] Examples of microelectrode arrays are provided in Bo Bi et al., *Building Addressable Libraries: The Use of "Safety-Catch" Linkers on Microelectrode Arrays*, 132 J. Am. Chem. Soc. 17,405 (2010); Bichlien H. Nguyen et al., *Microelectrode Arrays: A General Strategy for Using Oxidation Reactions To Site Selectively Modify Electrode Surfaces*, 30 Langmuir 2280 (2014); and U.S. Pat. Pub. No. 2020/1234567 filed on Jun. 7, 2019, with the title "Reversing Bias in Polymer Synthesis Electrode Array." One example of a microelectrode array and techniques for attaching polynucleotides to the surface of the array is provided in a Ryan D. Egeland & Edwin M. Southern, *Electrochemically Directed Synthesis of Oligonucleotides for DNA Microarray Fabrication*, 33(14) Nucleic Acids Res. e125 (2005).

[0037] The electrodes in a microelectrode array may be implemented with any known technology for creating microelectrodes such as complementary metal-oxide-semiconductor (CMOS) technology. CMOS may include metal-oxide-semiconductor field-effect transistors (MOSFETs) made through a triple-well process or by a silicon-on-insulator (SOI) process. A series of controllable gates/transistors implemented with CMOS circuits can be controlled to inject charge at any location on the surface of the microelectrode array. Each electrode in the microelectrode in the array may be independently addressed allowing the creation of arbitrary and variable voltage microenvironments across the surface of the microelectrode array.

[0038] High microelectrode density allows for fine-scale level control of the ionic environment at the surface of the microelectrode array. A microelectrode array may have a microelectrode density of approximately 1024 microelectrodes/cm.², approximately 12,544 microelectrodes/cm.², or a different density.

[0039] The array **100** may be covered with a plurality of spots **102(A)**, **102(B)**, . . . , **102(N)** at which initiators **104** are attached. Each of the initiators **104** is a single- or double-stranded polynucleotide strand. If double-stranded, the initiators **104** may have a 3' overhang, they may be blunt ended, or they may have a 3' recessed end. The length of an initiator **104** may be about 3-30 nucleotides, about 15-25 nucleotides, or about 20 nucleotides. The initiators **104** are not shown to scale. Because they are lengthened through repeated rounds of nucleotide addition, the initiators **104** may be referred to as growing (single-stranded) polynucleotide strands.

[0040] Although only three spots **102(A)**, **102(B)**, **102(N)** are shown in this illustrative

representation many thousands or hundreds of thousands of spots may be present on a typical array **100**. The size of a single spot **102** can be smaller than about 1 cm.^{sup.2}, smaller than about 1 mm.^{sup.2}, smaller than about 0.5 mm.^{sup.2}, smaller than about 100 μm .^{sup.2}, smaller than about 50 μm .^{sup.2}, smaller than about 1 μm .^{sup.2}, smaller than about 500 nm.^{sup.2}, or smaller than about 200 nm.^{sup.2}. Initiators **104** may also be present on the array **100** at locations other than the spots **102**.

[0041] The initiators **104** may be attached to the array **100** using any known technique for anchoring single-stranded DNA or RNA to a solid support such as techniques used in conventional solid-phase synthesis of polynucleotides or used for creation of DNA microarrays. For example, the initiators **104** may be spotted onto the array **100** by use of a robot to “print” pre-designed nucleotide sequences using fine-pointed pins, needles, or ink-jet printing onto a chemical matrix surface using surface engineering. Other methods employ photo-activated chemistry and masking to synthesize the initiators **104** one nucleotide at a time on the solid surface of the array **100** with a series of repeated steps to build up the initiators **104** at designated locations. In some implementations, the surface of the array **100** may be functionalized and the initiators **104** may be attached to the functional groups rather than directly to the array **100**.

[0042] All of the initiators **104** attached to the array **100** may have the same or approximately the same nucleotide sequence or one or more of the initiators **104** may have different sequences from the others. The sequence of any one or more of the initiators **104** may be a random sequence of nucleotides. The initiators **104** may also be constructed with non-random sequences such as, for example, sequences that are cleaved by a specific restriction endonuclease. Cleavage of the initiators **104** is one way to release completed polynucleotides from the surface of the array **100**. The sequences of the initiators **104** may also be designed or used as primer binding sites for subsequent amplification (e.g., polymerase-chain reaction (PCR) amplification) of fully synthesized polynucleotides.

[0043] Each spot **102** on the array **100** may contain many tens or hundreds of initiators **104** although for simplicity only three initiators **104** are shown on each spot **102** in this illustrative representation. Each initiator **104** attached to a single spot **102** is subject to the same spatially addressable control. Stated differently, any spatially addressable activation of a polymerase applied to the array **100** is performed at the resolution of individual spots **102**. However, the polynucleotides synthesized on the same spot **102** do not necessarily have the same nucleotide sequence because of the formation of variable length homopolymers.

[0044] The array **100** may be covered with a reaction reagent solution that contains inactive polymerase **106**. The reaction reagent solution is an aqueous solution that includes the polymerase and may also include buffers, salts, electrolytes, and the like. For example, the aqueous solution may include TdT buffer and a CoCl_2 solution to supply metal cofactors for the polymerase. There are many techniques for inactivating polymerase which are discussed in greater detail herein including blocking groups, unsuitable pH conditions, and enzyme inhibitors. Inactive polymerase **106** does not polymerize the addition of nucleotides to a growing polynucleotide strand or does so at a rate that is substantially less than the maximum rate of polymerization. For example, inactive polymerase **106** that still retains polymerase activity may perform nucleotide polymerization at rate that is less than 5%, 10%, 20%, 30%, or 40% of the maximum polymerization rate of the enzyme. Techniques for measuring DNA polymerase activity are known to those of ordinary skill in the art and described in WO 2010/036359.

[0045] Spatial activation of the polymerase creates active polymerase **108** at one or more of the spots **102** on the array **100**. Active polymerase **108** may be created by modifying inactive polymerase **106** or altering conditions such that the polymerase changes to an active state. For example, a blocking group attached to an inactive polymerase **106** may be released creating active polymerase **106**. The pH of the reaction reagent solution may be changed from a pH that prevents enzyme activity to an optimum pH for the polymerase that enables enzyme activity. Active

polymerase **108** catalyzes the addition of nucleotides to the 3'-end of a growing polynucleotide strand. Initially the growing polynucleotide strand is the initiator **104**. An active polymerase **108** may perform nucleotide polymerization at a rate that is at least 90%, 80%, 70%, or 60% of the maximum rate of polymerization for the polymerase.

[0046] In an alternative implementation (not shown in this figure) encapsulated enzyme inhibitors may be released from encapsulation thereby inhibiting polymerase activity. In this implementation, the array **100** is initially covered with active polymerase **108** that is changed to inactive polymerase **106** over one or more spots **102** where the enzyme inhibitors are released. The encapsulated enzyme inhibitors may be included in the reaction reagent solution.

[0047] In the illustrated representation, the inactive polymerase **106** located at spot A **102(A)** is changed to active polymerase **108**. This makes it possible for nucleotides to be incorporated onto the ends of the initiators **104** at spot A **102(A)** during the next synthesis cycle. Each added nucleotide extends the initiator **104** which becomes a growing polynucleotide strand. In a synthesis cycle, the surface of the array **100** may be flooded with a reaction reagent solution that contains inactive polymerase **106** and a selected species nucleotide **110** (i.e., a nucleotide having a specified base such as A, C, G, T, or U). Addition of only a single species of nucleotide limits the template-independent polymerase from randomly adding any nucleotide species. A nucleotide is a nucleoside linked to one or more phosphate groups. In some implementations, a nucleotide may be a deoxynucleoside triphosphate (dNTP) or a ribose triphosphate (NTP). Nucleotides are not limited to the canonical nucleotides but may also include nucleotide analogs.

[0048] In this example, a selected location **112** is defined by areas on the array **100** where the inactive polymerase **106** is changed to an active polymerase **108**. This representation shows the selected location **112** as spot A **102(A)**. The selected location **112** may be any one or more locations that are contiguous or separate on the surface of the array **100**. The selected location **112** may be a single spot **102**, a group of spots located adjacent to each other, or multiple disparate spots spread across the surface of the array **100**. In some implementations, the selected location **112** has an area less than 100 μm^2 , about 650 nm^2 , or about 200 nm^2 . The resolution or minimum size of the selected location **112** may be a single spot **102**. The spots **102** may be spaced apart from each other creating a buffer zone that functions to prevent nucleotide incorporation that spills over the edge of one spot **102(A)** from adding nucleotides to an adjacent spot **102(B)**.

[0049] Alternatively, if polymerase activity is controlled by enzyme inhibitors, release of enzyme inhibitors from encapsulation at the selected location **112** changes an active polymerase **108** to an inactive polymerase **106**. Thus, in this implementation the polymerase is inactive at the selected location **112** and active everywhere else.

[0050] The selected species of nucleotide **110** in this illustrative representation includes the base adenine (A) such as dATP or ATP. The selected species of nucleotide **110** may be provided as free nucleotides in solution. The free nucleotides are added to the 3'-ends of initiators **104** where there is active polymerase **108**. In an implementation, the selected species of nucleotide **110** may be an unmodified nucleotide that does not include a blocking group or any other modification. An unmodified nucleotide is a standard dNTP or NTP.

[0051] After the polymerase has had time to react with free nucleotides in solution (e.g., about 10, 20, 30, 40, 50 seconds, 1 minute, or 2 minutes) a wash step may be used to remove the polymerase and free nucleotides stopping extension of the initiators **104**. Depending on the length of time active polymerase **108** is in contact with the initiators **104**, multiple unmodified nucleotides may be added creating a homopolymer. Thus, the polynucleotide synthesis techniques of this disclosure may create block polymers in which the base-by-base sequence of polynucleotides on the same spot **102** is not always identical. However, the order of nucleotide blocks will be the same.

[0052] For example, a first cycle of synthesis may add from one to three adenine nucleotides to the 3' end of the initiators **104** at spot A **102(A)**. Additional nucleotides added during a single cycle of synthesis are referred to as extension region **114**. This variation arises from the ability of template-

independent polymerases to perform unregulated polymerization. Under a given set of reaction conditions the number of nucleotides added in an extension region **114** will vary with a distribution concentrated around a mean extension length. The reaction conditions include temperature, time, and the concentrations of the nucleotide, and concentration of the template-independent polymerase. The extension length may be tuned by adjusting the reaction time.

[0053] Thus, unless context indicates otherwise, “extension length” refers to the average extension length for a given set of reaction conditions. This variation in extension length for individual ones of the polynucleotides is the reason why a population of polynucleotides at the same spot **102** may have different sequences. Due to the presence of homopolymers it may not be possible to synthesize polynucleotides with specific base-by-base sequences using unmodified nucleotides.

[0054] However, polynucleotides that have a specified order of nucleotide bases even if the precise number of nucleotides is not controllable have uses in applications such as digital data storage. Data may be included in nucleotide transitions rather than by the absolute sequence. Techniques for encoding digital data using nucleotide transitions in polynucleotides with homopolymers are discussed in Henry H. Li et al., *Terminator-free template-independent enzymatic DNA synthesis for Digital information storage*. 10:2383 Nat. Comm. (2019), WO 2017/176541 A1, and U.S. Pat. Pub. No. US2020/0058216 filed on Aug. 16, 2019, with the title “Regulation of Polymerase Using Cofactor Oxidation States.”

[0055] In some implementations, the selected species of nucleotide **110** may be modified such as by addition of a 3' protecting group. Techniques for creating modified nucleotides that include a 3' protecting group are known to those of skill in the art. Illustrative 3' protecting groups are described in U.S. Pat. Pub. No. US2021/0058216 filed on May 28, 2020, with the title “De novo polynucleotide synthesis with substrate-bound polymerase.”

[0056] The selected species of nucleotide **110** is added to the 3'-end of an initiator **104** and the 3' protecting group remains preventing further nucleotide addition. Thus, by use of modified nucleotides with protecting groups the synthesis techniques of this disclosure can be used to create polynucleotides with specific base-by-base sequences. Once the active polymerase **108** is removed either by a wash step or by conversion to inactive polymerase **106** the blocking groups on the ends of the initiators **104** are removed and the cycle can be repeated.

[0057] However, if homopolymers can be tolerated in the final polynucleotides, use of unmodified nucleotides provides a cost benefit because unmodified nucleotides are less expensive than modified nucleotides. For example, dNTPs with protecting groups such as CleanAmp® dNTPs available from TriLink® Biotechnologies cost approximately 2.5 times more than equivalent unprotected dNTPs.

[0058] FIG. 2 shows use of a blocking group **200** reversibly attached by a His-tag sequence **204** as one technique to regulate polymerase activity. When the blocking group **200** is attached it sterically hinders activity of the polymerase. The polymerase is then unable to add nucleotides to the end of a growing polynucleotide strand or the rate of addition is greatly slowed. The blocking group **200** may include a ligand **202** that is complexed to a His-tag sequence **204** on the polymerase. In an implementation, the blocking groups **200** are covalently modified with ligands **202**. The polymerase is an inactive polymerase **106** when the blocking group **200** is attached and becomes an active polymerase **108** when the blocking group is detached.

[0059] A polymerase, such as TdT, may be covalently modified with a polyhistidine tag or His-tag sequence **204**. Polymerases that include a His-tag sequence **204** may be created by well-known protein production process including expression in *E. coli* or yeast. The His-tag sequence **204** may be added to the N-terminal or the C-terminal of the polymerase. Modifications of TdT to include a His-tag for protein purification are described in U.S. Pat. Nos. 10,435,676, 10,059,929, U.S. Pat. Pub. No. 2019/0360013, and Sebastian Barthel et al., *Enhancing Terminal Deoxynucleotidyl Transferase Activity on Substrates with 3' Terminal Structures for Enzymatic De Novo DNA Synthesis*, 11 Genes 102 (2020). Proteins may also be synthesized with His-tags using techniques

such as those described in Chih-Chi Chu and Scott K. Silverman, *Assessing histidine tags for recruiting deoxyribozymes to catalyze peptide and protein modification reactions*, 14 Org. Biomol. Chem 4697 (2016).

[0060] A His-tag sequence **204** binds metal centers such as certain divalent metal cations **206**. The behavior of polyhistidine tags is known to those of ordinary skill in the art and discussed in Richard J. Sundberg and R. Bruce Martin, *Interactions of histidine and other imidazole derivatives with transition metal ions in chemical and biological systems*, 74(4) Chem Rev. 471 (1974). The divalent metal cation **206** may be a transition metal. The divalent metal cation **206** may be nickel (Ni.sup.2+), cobalt (Co.sup.2+), copper (Cu.sup.2+), or zinc (Zn.sup.2+). In one implementation, the divalent metal cation **206** is Cu.sup.2+. The divalent metal cation **206** may be provided in solution as a salt. For example, Cu.sup.2+ may be provided by any water-soluble copper (II) salt such as cupric chloride, cupric acetate, or cupric sulfate.

[0061] The polymerase that includes a His-tag sequence **204** may be initially complexed with a blocking group **200** through the His-tag sequence **204**. The blocking group **200** may be an inactive protein or other polymer of sufficient size to prevent, or greatly reduce, the activity of the polymerase. In an implementation, the blocking group **200** has a molecular weight that is between about 0.5-10× the molecular weight of the polymerase. In an implementation, the blocking group **200** has a weight that is about the same as the polymerase. The template-independent polymerase TdT weighs about 58 kD and includes about 500 amino acids. Thus, in some implementations the blocking group **200** may have a molecular weight of about 50-600 kD. If the blocking group **200** is a protein, its length may be about 50-600 amino acids. In an implementation, the molecular weight of the blocking group may be about 58 kD. In an implementation, if the blocking group **200** is a protein its length may be about 500 amino acids.

[0062] The blocking group **200** may be covalently modified with a ligand **202** that complexes with the divalent metal cation **206** and the His-tag sequence **204**. The ligand **202** may be, for example, a poly(nitrilotriacetic acid) tag (NTA) or an iminodiacetic acid tag (IDA). Techniques for attaching an NTA tag or an IDA tag to a protein or polymer are known to those of skill in the art. Illustrative techniques for attaching NTA tags to other molecules are described in Russell P. Goodman et al., *A Facile Method for Reversibly Linking a Recombinant Protein to DNA*, 10(9) ChemBioChem 1551 (2009).

[0063] The complex between the His-tag sequence **204** and the ligand **202** is broken if the divalent metal cation **206** is changed to a different oxidation state. For example, the divalent metal cation **206** may be changed to a monovalent metal cation **208**. As a further example, Cu.sup.2+ may be reduced to Cu.sup.+ . The blocking group **200** will then disassociate from the polymerase leaving active polymerase **108**. The process is reversible. If a suitable divalent metal cation **206** becomes available, the blocking group **200** will reattach via the His-tag sequence **204**—ligand **202** complex and make the polymerase inactive.

[0064] Reduction or oxidation of the divalent metal cation **206** may be achieved by any number of mechanisms. Activation of an electrode such as in a microelectrode array can create a localized change in redox conditions causing the reduction or oxidation of a divalent metal cation **206**. For example, a potential of about +0.16 V vs. standard hydrogen electrode (SHE) in an aqueous solution can reduce Cu.sup.2+ to Cu.sup.+ . A potential of about -0.25 V vs. SHE in an aqueous solution can reduce Ni.sup.2+ to Ni.sup.0.

[0065] The redox reaction may be initiated directly or indirectly at an electrode surface. At the electrode surface, reduction or oxidation will take place using electron transfer directly at the electrode or mediated by the redox of a mediator. Redox mediators are chemicals with electrochemical activity. In a bioelectrocatalysis process, mediators may exchange electrons with fuels or oxidants at the reaction sites of the biocatalysts and then diffuse to the surface of electrode and exchange electrons there. Use of mediators may also reduce the required electrode potential which in turn reduces the energy needed to change the divalent metal cation **206** into a monovalent

metal cation **208** or other oxidation state.

[0066] Additionally, application of a redox reagent by a fluid deposition instrument such as a chemical inkjet printer may be used to reduce or oxidize the divalent metal cation **206** at selected locations on the surface of an array. The redox reagent contributes or receives electrons from the divalent metal cation **206** changing it to a different oxidation state. For example, the reducing agent such as ascorbic acid (e.g., provided as a salt such as sodium ascorbate), citric acid, sodium hypophosphate, hydrazine, including their salts, or similar may be added to reduce Cu.sup.2+ to Cu.sup.+ . The oxygen may be present as dissolved atmospheric oxygen in the reaction reagent solution or may be provided such as the addition of hydrogen peroxide. The volume of redox agent added is determined by the capabilities of the fluid deposition instrument and the size of area over which the polymerase is to be activated. The concentration of the redox reagent is determined in part by the volume used and the concentration of divalent metal cations **206** in solution. For example, the redox reagent may be added such that the at the selected location there is about 0.1-10 mole equivalents to the divalent metal cations **206**. In an implementation, the redox reagent and the divalent metal cations **206** are present in about equimolar amounts at the selected location.

[0067] A reduction or oxidation of the divalent metal cation **206** may also be induced by exciting a light-activated photoredox catalyst. The photoredox catalyst may be, for example, a metal polypyridyl complex such as an iridium polypyridyl complex. Any other type of suitable photoredox catalyst may also be used such as, for example, organic photoredox catalysts some of which are described in Nathan A. Romero and David A. Nicewicz, *Organic Photoredox Catalysis*, 116(17) Chem. Rev. 10075 (2016). Light that excites the photoredox catalyst may be directed onto selected locations on an array to provide spatially addressable activation of the polymerase. The photoredox catalysts may be included in the reaction reagent solution. The photoredox catalyst may be added such that the at the selected location there is about 0.1-1 mole equivalents to the divalent metal cations **206**.

[0068] The change from inactive polymerase **106** to active polymerase **108** is confined to a selected location on the surface of the array by an excess of divalent metal cations **206** and optionally also an excess of blocking groups **200** in the reaction reagent solution. As active polymerase **108** diffuses away from the selected location, the excess divalent metal cations **206** and blocking groups **200** will cause a blocking group **200** to reattach changing the active polymerase **108** back into inactive polymerase **106**. This provides spatial control of polymerase activity. For example, using this technique active polymerase **108** may be confined to one or more of the spots **102** shown in FIG. 1.

[0069] Additionally, if an electrode is used to change the oxidation state of the divalent metal cation **206**, the excess of divalent metal cations **206** will cause the blocking groups **200** to reattach once the current is turned off thereby stopping polynucleotide extension at the selected location. The duration of electrode activation may be used to limit the time that active polymerase **108** is available and thus regulate the extension length.

[0070] The divalent metal cations **206** that complex with the His-tag sequence **204** are not the only divalent metal cations in the reaction reagent solution. The polymerase complexes with a cofactor that is also a divalent metal cation. A template-independent polymerase such as TdT is able to use a variety of divalent metal cations such as Co.sup.2+, Mn.sup.2+, Zn.sup.2+ and Mg.sup.2+. However, due to the difference in chelating groups and steric environments, a redox potential sufficient to reduce (or oxidize) the divalent metal cation **206** chelated to a His-tag sequence **204** is lower than the redox potential needed to change the oxidation state of a metal cofactor complexed with an enzyme. See Scott E. Waters et al., *Effect of Chelation on Iron-Chromium Redox Flow Batteries*, 5 ACS Energy Lett. 1758 (2020) and Yasukazu Yoshida et al., *Effects of metal chelating agents on the oxidation of lipids induced by copper and iron*, 1210 Biochimica et Biophysica Acta 81 (1993). Thus, the same metal ion may be in the His-tag linkage and as an enzyme cofactor by selecting a redox potential sufficient to cause disassociation of the His-tag linkage without

changing the oxidation state of the metal cofactor.

[0071] In an implementation, the metal cofactor and the divalent metal cation **206** may be selected so that they are different metals. For example, if a cobalt (Co.sup.2+) metal cofactor is provided for the polymerase, then the divalent metal cation **206** complexed with the His-tag sequence **204** is a different metal cation such as copper (Cu.sup.2+). The reduction potential of Co.sup.2+ to Co is -0.28 V vs. SHE as compared to +0.16 V vs. SHE for Cu.sup.2+ to Cu.sup.+ . Thus, even if both metal ions were in the same environment, the voltage applied by an electrode or the amount and strength of redox reagent or photoredox catalyst that reduce Cu.sup.2+ will not change the oxidation state of Co.sup.2+.

[0072] FIG. **3** shows an array **100** for solid-phase polynucleotide synthesis on which spatial control of polymerase activity is regulated by pH. The array **100** is coated with a plurality of initiators **104**. The array **100** may initially be contacted with a reaction reagent solution buffered to an unsuitable pH **300** and inactive polymerase **106**. The unsuitable pH is sufficiently acidic or basic to change the conformation of the polymerase in a way that decreases or eliminates enzyme activity. For example, an unsuitable pH for TdT may be below about 4 or above about 10. A selected species of nucleotide **110** is added either together with the reaction reagent solution or separately. Due to the inactivating effects of the unsuitable pH **300**, polymerization does not occur or is very slow. Thus, inactive polymerase **106** is present where the pH is an unsuitable pH **300**.

[0073] The pH of solution contacting the array **100** is changed to an optimum pH **302** for the polymerase at the selected location **112**. The selected location **112** may be any shape and may cover any portion of the array **100**. In an implementation, the selected location **112** may be one or more spots **102** as illustrated in FIG. **1**. Optimum ranges of pH for enzyme activity are known to those of ordinary skill in the art and may be found in F. J. Bollum, In: *The Enzymes*, Boyer, P.D., ed, Academic Press, New York, 145 (1974). The optimum pH **302** for TdT is about 6-8 pH.

[0074] In one implementation, the initial pH of the reaction reagent solution is an unsuitable pH **300** that is higher than the optimum pH **302**. The pH is then reduced at the selected location **112** by addition of acid with a targeted deposition instrument. Alternatively, light may be used to activate photoacid generators at the selected location **112** to reduce the pH to the optimum pH **302** for the polymerase. As used herein, photoacid generators include both photoacid generators (irreversible proton photon dissociation) and photoacids (reversibly recombine when exposure to the light source ceases). Many suitable photoacid generators are known including phenols, naphthols, and pyrenes (see Heike Kagel et al., Photoacids in biochemical applications, 4 J. Cell. Biotech. 23 (2018)). Suitable photoacid generators include photosensitive-2-nitrobenzyl esters (see Pawel J. Serafinowski and Peter B. Garland, *Novel Photoacid Generators for Photodirected Oligonucleotide Synthesis*, 125(4) J. Am. Chem. Soc. 962 (2003)). Suitable water-soluble photoacid generators are also known (see Zheng Shi et al., *Long-Lived Photoacid Based upon a Photochromic Reaction*, 133(37) J. Am. Chem. Soc. 14699 (2011)).

[0075] In one implementation, the initial pH of the reaction reagent solution is an unsuitable pH **300** that is lower than the optimum pH **302**. The pH is then increased at the selected location **112** by addition of base with a targeted deposition instrument. Alternatively, light may be used to activate photobase generators at the selected location **112** to increase the pH to the optimum pH **302** for the polymerase. Examples of suitable photobase generators tetraphenylborate salt of bicyclic guanidine base, 1,5,7-triaza-bicyclo[4.4.0]dec-5-ene (TBD) (Cheng B. Cao, et al., *Photo-induced crosslinking of water-soluble polymers with a new photobase generator*, 51(18) Polymer, 4058 (2010) and WPBG-266 which is water-soluble and available from FujiFilm Wako Pure Chemical Corporation.

[0076] Electrodes, such as a microelectrode array, may also be used to change the pH of an aqueous solution. The pH of water can be modified electrochemically over a wide range. The maximum concentration limits of base and acid that can be produced in a given water sample limit the range of pH changes. These limits are determined by the kinds and concentrations of all

elements dissolved in the water. It is possible, using electricity, to produce water with any content of base or acid within these limits. Electrochemical pH control is based on electrical decomposition of water in an electrolytic cell divided by an ion-exchange membrane or diaphragm into anode and cathode compartments. The minimum (theoretical) voltage necessary to decompose water is 1.23 V at 25° C. Actual decomposition voltage is higher because of the irreversible nature of electrodes. A visible evolution of gases (O₂ and H₂) commences at 1.7 V. See Albert Regner, *Electrochemical processes in chemical industries*. Artia, Prague: 198-212 (1957). Usually either the anodic or the cathodic reaction is used for pH control, while the complementary reaction may be undesired in some practical applications. An appropriate voltage is applied to a subset of electrodes at the selected location **112** so as to generate an optimum pH **302** in local environment immediately around the electrodes.

[0077] The pH change is confined to the selected location **112** by buffer in the solution (e.g., reaction reagent solution) in contact with the array **100**. The strength of the buffer affects the degree of confinement. Active polymerase **108** incorporates the selected species of nucleotide **110** on to the ends of the initiators **104** at the selected location **112**. If the selected species of nucleotide **110** is an unmodified nucleotide, due to its ability to perform unregulated addition the polymerase may add multiple nucleotides to the ends of the initiators **104**.

[0078] FIG. **4** shows an array **100** for solid-phase polynucleotide synthesis on which spatial control of polymerase activity is regulated by release of encapsulated enzyme inhibitors **400**. Initiators **104** are attached to the surface of the array **100** as in the other implementations. A solution contacting the array, e.g., the reaction reagent solution, contains encapsulated enzyme inhibitors **400**. The enzyme inhibitor **402** may be any type of inhibitor that, for a particular polymerase, changes it from an active polymerase **108** to an inactive polymerase **106**. Numerous enzymatic inhibitors **402** are known to those of ordinary skill in the art. Enzymatic inhibitors **402** that are known to inhibit TdT include ATP and its dinucleotide analogs. See Pandey, V.N., Amrute, S.B., Satav, J.G. and Modak, M.J., *Inhibition of terminal deoxynucleotidyl transferase by adenine dinucleotides*, FEBS Letters, 213 (1987).

[0079] The encapsulate **404** cages the enzyme inhibitor **402** and prevents its inhibitory action. Many different techniques are known to those of ordinary skill in the art for pH-dependent encapsulation of a drug or other compound. There are many established systems for encapsulating drugs including hydrogels, nanoparticles, microspheres, acid-cleavable micelles, and amphiphilic assemblies. Some examples are described in Lin Liu et al., pH-Responsive carriers for oral drug delivery: challenges and opportunities of current platforms, 24:1 Drug Delivery, 569 (2017). Any of these techniques, or others, may be adapted to encapsulate an enzyme inhibitor such as ATP. For example, ATP has been encapsulated in liposomes as described in W. Liang, et al., Encapsulation of ATP into liposomes by different methods: optimization of the procedure, 21(3) J. Microencapsulation 251 (2004).

[0080] The pH at a selected location **112** on the array **100**, for example spot B **102(B)**, is changed thereby causing the encapsulate **404** to release the enzyme inhibitor **402** and create inactive polymerase **106**. The pH at the selected location **112** may be changed by any of the techniques mentioned previously. The specific pH change required depends on the encapsulate **404** and can be readily identified by one of ordinary skill in the art. The pH changes to trigger opening of encapsulates **404** may occur at any pH even an unsuitable pH for the polymerase because the unsuitable pH would have synergistic effect with the enzyme inhibitor **402**.

[0081] With the use of encapsulated enzyme inhibitors **400**, polymerase activity is inhibited or stopped at the selected location **112**. Thus, prior to release of the enzyme inhibitors **402**, active polymerase **108** may be present across the entire surface of the array **100**. To prevent undifferentiated incorporation of nucleotides, polymerase activity may be suppressed across the entire array **100** prior to release of the enzyme inhibitors **402**. One technique for reducing polymerase activity is cooling to a temperature that prevents or greatly slows activity of the

polymerase.

[0082] The temperature of the aqueous solution contacting the array **100** may be cooled to a temperature a few degrees above freezing. The specific freezing temperature will depend on the concentration and type of salts and buffers in solution. For example, a reaction reagent solution containing the polymerase may be cooled to about 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10° C. prior to mixing with the selected species of nucleotide. Then, after a change in pH has released the enzyme inhibitors **402** at the selected location **112**, the temperature is increased to a temperature at which the polymerase is active.

[0083] The temperature of all the solution in contact with the array **100** may be increased by use of a heater such as a resistor. If electrodes are used to change the pH and release the enzyme inhibitors **402**, the current passing through the reaction reagent solution alone may be sufficient to increase the temperature at the selected location **112**.

[0084] Even when heated, the polymerase remains inactive polymerase **106** at the locations where the enzyme inhibitor **402** is released. Thus, in this implementation, polynucleotide extension is inhibited at the selected location **112** and the initiators **104** are extended at locations other than the selected location **112**. Inhibition may not be complete. There may be some polymerization at the selected location **112**. For example, active polymerase **108** may be in contact with the entire array **100** for a brief period of time during each synthesis cycle. The enzyme inhibitors **402** may also diffuse away from the selected location **112** and reduce polymerization at off target areas. However, differences in the amount of polymerization, such as the length of extension regions **114**, will be detectably different at the selected location **112** and other areas of the array **100**.

[0085] Polynucleotides that have the same sequence of nucleotide blocks but differ in the length of the blocks are useful for applications such as encoding information for digital data storage.

Illustrative Process

[0086] FIG. 5 shows process **500** for de novo synthesis of polynucleotides by spatial control of polymerase activity. This process **500** may be implemented, for example, using any of the processes, techniques, or reactions, shown in FIGS. 1-4 or the device shown in FIG. 6.

[0087] At operation **502**, an array is incubated with a reaction reagent solution that includes template-independent polymerase and a selected species of nucleotide. The array may be made out of silicon dioxide, glass, an insoluble polymer, or other material. In an implementation, the array is a microelectrode array. In some implementations, the template-independent polymerase is TdT.

[0088] A plurality of initiators are attached to the surface of the array. The array may be covered with many thousands or millions of separate initiators. The initiators are single-stranded nucleotides with a length of between about 3-30 bases. Each of the initiators may be identical having the same length and nucleotide sequence. However, there may also be variation among the initiators in terms of length as well as sequence. In some implementations, the sequences of the initiators may include a cut site for restriction enzymes or other nucleases to cleave the polynucleotides from the surface of the array. In some implementation, the initiators may serve as primer binding sites for subsequent amplification of the polynucleotides synthesized on the array.

[0089] The template-independent polymerase uses the initiators as a starting point for addition of additional nucleotides to the 3' terminal nucleotide at the end of each initiator. The initiators may be attached to the array by any known or later developed technique for anchoring single-stranded DNA or RNA to a solid support. Example techniques include those used in conventional solid-phase synthesis of DNA and used for creation of DNA microarrays.

[0090] The reaction reagent solution may be delivered to a reaction chamber that contains the array. The reaction reagent solution may be added to the reaction chamber by a manual technique such as pipetting. The reaction reagent solution may be added to the reaction chamber by an automated or mechanized system such as via a fluid delivery pathway. The reaction reagent solution includes a substrate-independent polymerase in an appropriate buffer or salt solution.

[0091] In one implementation, only unmodified nucleotides are incubated with the reaction reagent

solution. For example, the selected species nucleotide may be one of deoxyadenosine triphosphate (dATP), deoxycytidine triphosphate (dCTP), deoxyguanosine triphosphate (dGTP), deoxythymidine triphosphate (dTTP), adenosine triphosphate (ATP), cytidine triphosphate (CTP), guanosine triphosphate (GTP), or uridine triphosphate (UTP).

[0092] Incubation continues for a length of time referred to as a reaction time. The reaction time may be any length of time sufficient for polymerization to occur. Until activity of the template-independent polymerase is stopped, increased reaction time increases the extension length. For example, the reaction time may be about 10, 20, 30, 40, 50 seconds, 1 minute, or 2 minutes.

[0093] Activity of the template-independent polymerase is stopped at the end of the reaction time. The length of time until activity of the template-independent polymerase is stopped may define the reaction time. In an implementation, activity of the template-independent polymerase is stopped by a wash step that removes the template-independent polymerase and the free nucleotides from the surface of the array.

[0094] At operation **504**, the redox-pH condition is altered at a selected location on the array. A change in redox-pH conditions includes a change in redox conditions or a change in pH conditions. Altering the redox-pH condition affects the activity of the polymerase. The polymerase may be changed from an inactive polymerase that does not perform appreciable nucleotide polymerization to an active polymerase that adds nucleotides to the end of a growing polynucleotide strand. An inactive polymerase may have a rate of polymerization that is decreased to zero or to a level that is statistically distinguishable from active polymerase. The rate of polymerization may be effectively zero if the rate is sufficiently slow such that the polymerase does not incorporate nucleotides during the time when the array is in contact with the reaction reagent solution (e.g., one minute or less). In an implementation, changing the activity of the template-independent polymerase means increasing or decreasing the rate of polymerization activity by at least one order of magnitude.

[0095] The selected location may be any one or more locations that are contiguous or separate on the surface of the array. The selected location on the array may be one or more spots that each contain multiple individual initiators such as the spots **102** illustrated in FIG. **1**. The selected location may be a single spot, a group of spots located adjacent to each other, or multiple disparate spots spread across the surface of the array in any pattern. The selected location may be changed one or more times during the synthesis of polynucleotides on the array.

[0096] Any one of multiple different techniques may be selected to alter the redox-pH conditions on the array. The redox-pH conditions may be altered in a spatially addressable way by activating electrodes on a microelectrode array; adding a redox reagent, acid, or base with a targeted fluid deposition instrument; or exposing a photoredox catalyst, a photoacid, or a photobase in solution to a light source. The light source may be directed onto the array in a site-specific manner by a photomask, digital micromirror device (DMD), or other type of optoelectronics. The photoredox catalyst, photoacid, or photobase may be included in the reaction reagent solution.

[0097] In one implementation, a change in redox conditions is used to activate the polymerase at the selected location by causing the release of a blocking group that prevents polymerase activity. Activating the polymerase results in an increase in the rate of polymerization activity. A blocking group may be attached to the polymerase by a His-tag complexed with divalent metal cations. A change in redox conditions at the selected location may cause a localized change in the oxidation state of the metal cations leading to release of the blocking group. The change in redox conditions may be induced by one or more electrodes of a microelectrode array applying a potential of about +0.2, +0.3, or +0.4 V vs SHE. The change in redox conditions may be induced by a redox reagent applied to the surface of the array by a targeted fluid deposition instrument. The redox reagent may be, for example, ascorbic acid, citric acid, sodium hypophosphate, hydrazine, or similar. The change redox conditions may be induced by photo activation of a photoredox catalyst. The photoredox catalyst may be, for example, a metal polypyridyl complex.

[0098] In one implementation, a change in pH conditions is used to activate the polymerase at the

selected location by changing the pH to an optimum pH for the enzyme. The pH may be maintained across substantially the entire surface of the array at an unsuitable pH to prevent enzyme activity. The pH is then raised or lowered at the selected location into the range of optimum pH for the polymerase. The pH may be changed by activation of electrodes in a microelectrode array, by addition of an acid or base from a targeted fluid deposition instrument, or by activation of a photoacid or photobase upon exposure to a light source.

[0099] In one implementation, the change in pH conditions is used to inactivate the polymerase at the selected location by releasing enzyme inhibitors from encapsulation that inhibits activity of the polymerase. Inactivating the polymerase results in a decrease in the rate of polymerization activity. Encapsulates that prevent enzyme inhibitors from interacting with the polymerase may respond to changes in pH. A sufficient change in the pH for the specific type of encapsulate will cause it to release enzyme inhibitors. The localized pH at a selected location may be changed by any of the techniques described above. Thus, there will be inactive polymerase at the selected location and there will be active polymerase at other locations on the array. The encapsulated enzyme inhibitors may be provided in the reaction reagent solution.

[0100] Regulation of enzyme activity by encapsulated enzyme inhibitors may be combined with regulation through control of temperature. Temperature may be used to prevent or slow polymerase activity until the enzyme inhibitors are released from encapsulation. For example, the temperature of the reaction reagent solution may be maintained at a low temperature that decreases polymerase activity until the enzyme inhibitors are released and then increased to a suitable temperature for the polymerase. This can prevent nucleotide addition from occurring over the whole array prior to release of the enzyme inhibitors.

[0101] At operation **506**, the array is contacted with a wash solution. The wash solution may be flowed across the entire array displacing the polymerase and any free nucleotides thereby stopping further extension of the growing polynucleotide strands. Washing between separate cycles of synthesis also prevents contamination of free nucleotides from a previous cycle. The wash solution may be water such as DI (deionized) water. The wash solution may be an aqueous solution that contains at least one of a salt or a buffer. The buffer may be any one of a number of aqueous buffers known to those of ordinary skill in the art that are compatible with polymerases and single-stranded nucleotides such as phosphate-buffered saline (PBS) or tris-buffered saline (TBS). The wash solution may also include a denaturing agent such as a surfactant (e.g., 1% sodium dodecyl sulfate) or a protease (e.g., Proteinase K) to inactivate enzymes on the array.

[0102] At operation **508**, it is determined if polynucleotide synthesis is complete. If all nucleotides needed to create the specified sequences of the polynucleotides being synthesized on the array have been added, then polynucleotides synthesis is complete. If complete, process **500** proceeds along the “yes” path to operation **510**.

[0103] If, however, polynucleotide synthesis is not complete, process **500** proceeds along “no” path and returns to **502** where the array is again incubated with a reaction reagent solution and selected species of nucleotide. In a subsequent synthesis cycle the selected location and the species of nucleotide may both be independently changed. Process **500** may be iteratively repeated such that both the selected location and the selected nucleotide are changed at least once between iterations. This allows for the parallel synthesis of multiple polynucleotides each with a different sequence on the surface of a single array.

[0104] At operation **510**, the polynucleotides are separated from the array. If the polynucleotides are attached to the array by linkers, cleavage of the linkers may release the polynucleotides. The initiators used to start growth of polynucleotides on the array may be cleaved by restriction enzyme digests. Other techniques for separating polynucleotides from a solid substrate following solid-phase synthesis are known to those of ordinary skill in the art. Any suitable technique may be used. The polynucleotides may be collected and stored or processed further such as by amplification with polymerase chain reaction (PCR).

Illustrative System and Device

[0105] FIG. 6 is an illustrative system for implementing aspects of this disclosure. The system includes a device for de novo synthesis of polynucleotides that may be referred to as a polynucleotide synthesizer **600**. The system may also include a computing device **602**. The computing device **602** includes at least one or more processing units and memory such as random-access memory (“RAM”) and/or read-only memory (“ROM”) communicatively coupled to the processing units. The computing device **602** may also include a mass storage device configured to store files, documents, and data such as, for example, sequence data that is provided to the polynucleotide synthesizer **600** in the form of instructions. The computing device **602** may be implemented as any type of conventional computing device such as a desktop computer, a laptop computer, a server, a hand-held device, or the like. In an implementation, the computing device **602** may be a part of the polynucleotide synthesizer **600** rather than a separate device.

[0106] The computing device **602** may include a polynucleotide synthesizer control module **604**. The polynucleotide synthesizer control module **604** provides instructions **606** that can control operation of polynucleotide synthesizer **600**. For example, the instructions **606** may communicate to the polynucleotide synthesizer **600** base sequences of polynucleotides for synthesis.

[0107] The polynucleotide synthesizer **600** is a device that performs automated solid-phase synthesis of on an array **100**. The array **100** may be located within a reaction chamber **610** configured to maintain an aqueous solution such as a reaction reagent solution **612** in contact with the surface of the array **100**. The polynucleotide synthesizer **600** may also include a heater to control the temperature of the aqueous solution in the reaction chamber **610**. The polynucleotide synthesizer **600** may also include a cooling device such as a fan or thermoelectric cooler (e.g., Peltier device) to lower the temperature of the aqueous solution in the reaction chamber **610**. As described above, the temperature may be lowered when introducing polymerase in order to suppress activity until enzyme inhibitors are released.

[0108] The array **100** may be formed from one or more of silicon dioxide, glass, an insoluble polymer, a non-reactive metal such as gold, silver, or platinum, or other material. The array **100** may be an electrochemically inert surface or it may include a plurality of spatially addressable microelectrodes. Thus, in an implementation, the array **100** may be a microelectrode array with individually addressable electrodes.

[0109] The polynucleotide synthesizer **600** may also include storage tanks, bottles, vials, or other containers or receptacles for storing solutions and reagents used in the synthesis of polynucleotides. One such receptacle may contain the reaction reagent solution **612**. The reaction reagent solution **612** is an aqueous solution that contains a template-independent polymerase, metal cofactors for the polymerase, and at least one of a salt or buffer.

[0110] The buffer may be any one of a number of known aqueous buffers that are compatible with polymerases such as, for example, PBS. PBS is a water-based salt solution containing disodium hydrogen phosphate, sodium chloride and, in some formulations, may also include one or more of potassium chloride and potassium dihydrogen phosphate. Other examples of aqueous buffers known to those of ordinary skill in the art include HEPES, MOPS, PBST, TAE, TBE, TBST, TE, and TEN. See Vincent S. Stoll & John S. Blanchard, *Buffers: Principles and Practice*, 182 Meth. Enzoml., 24 (1990).

[0111] Nucleotides **614** may be stored separately in different receptacles. Each species of nucleotide may be stored in a separate receptacle. If synthesizing DNA, the nucleotides **614** may be dNTPs that include one of the natural bases adenine (A), guanine (G), cytosine (C), or thymine (T). If synthesizing RNA, the nucleotides **614** may be NTPs that include one of the natural bases adenine (A), guanine (G), cytosine (C), or uracil (U). In an implementation the nucleotides are unmodified. In an implementation, the nucleotides are modified to include a blocking group such as a 3'-OH blocking group.

[0112] Although four different types of nucleotides **614** are illustrated in FIG. 6, the polynucleotide

synthesizer **600** may include fewer types (e.g., omit one of the standard nucleotides) or more types (e.g., include one or more artificial nucleotides). Only one species of nucleotide is provided during each cycle of synthesis to control which nucleotide is next incorporated by the template-independent polymerase into the polynucleotides. However, different ones of the available nucleotides **614** may be introduced during different cycles of synthesis to create a plurality of polynucleotides at different locations each with a different nucleotide sequence.

[0113] Both the reaction reagent solution **612** and the nucleotides **614** may be brought into contact with the array **100** by a fluid delivery pathway **616** that has a fluid connection with the reaction chamber **610**. A first fluid delivery pathway **616(A)** may deliver reaction reagent solution **612** to the reaction chamber **610**. A second fluid delivery pathway **616(B)** may deliver a selected species of nucleotide **614** to the reaction chamber **610**. A third fluid delivery pathway **616(C)**, fourth fluid delivery pathway **616(D)**, fifth fluid delivery pathway **616(E)** and so on may deliver other species of nucleotides **614** to the reaction chamber **610**. The fluid delivery pathways **616** may be implemented by one or more of tubes and pumps, microfluidics, laboratory robotics, manual pipetting, or other techniques that move controlled volumes of fluids from one location to another.

[0114] The polynucleotide synthesizer **600** may also include a wash solution **618**. The wash solution **618** may be water (e.g., DI (deionized) water) or an aqueous solution that contains at least one of a salt or a buffer. The salt or the buffer may be the same as the salt or buffer used in the reaction reagent solution **612**. Alternatively, the salt or the buffer may be a different salt or buffer that is suitable for washing polynucleotides such as PBS or TBS. The wash solution **618** is flowed into the reaction chamber **610** through a fluid delivery pathway **616(F)**. The wash solution **618** displaces the template-independent polymerase and any free nucleotides **614** in the reaction chamber **610**. By removing any free nucleotides **614**, the subsequent cycle of polymerization can introduce a different species of nucleotide without contamination from the previous cycle. Multiple cycles of addition of the same nucleotide **614** are possible and may each be followed by a wash step even though the same species of nucleotide is being added.

[0115] One or more other reagents **620** may also be included in the polynucleotide synthesizer **600** and brought into contact with the array **100** through a fluid delivery pathway **616(G)**. The other reagents **620** may include, for example, a deblocking agent to deblock protected nucleotides or a supporting electrolyte.

[0116] As discussed above, there are multiple different devices and techniques for controlling polymerase activity on the surface of the array **100** by changing redox-pH conditions. All of these different devices and techniques are referred to collectively as a redox-pH control mechanism **622**. Polymerization is promoted at one or more selected locations on the array **100** by increasing the rate of nucleotide polymerization from zero or a negligible level to a level at which the polymerase incorporates free nucleotides onto the end of a growing polynucleotide. This may be thought of as “activating” the polymerase. Polymerization is inhibited at one or more selected locations on the surface of the array **100** by reducing the rate of nucleotide polymerization to zero or a negligible level. This may be thought of as “inactivating” polymerization. The spatial pattern of polymerase activation and inactivation causes location-specific polymerization and enables the creation of polynucleotides with different sequences on the array **100**.

[0117] The redox-pH control mechanism **622** may be any of a microelectrode array **624**, a targeted fluid deposition instrument **626**, or a light source **628**. The redox-pH control mechanism **622** may be controlled by control circuitry **630**. The control circuitry **630** may be implemented as any type of circuitry suitable for controlling hardware devices such as a printed circuit board, microcontroller, a programmable logic controller (PLC), or the like. The control circuitry **630** receives and acts on the instructions **606** provided by the polynucleotide synthesizer control module **604**. Thus, the control circuitry **630**, through the redox-pH control mechanism **622**, can control where nucleotide polymerization occurs on the array **100**. The control circuitry **630** may also control the fluid delivery pathways **616**. For example, the control circuitry **630** may

sequentially open the various fluid delivery pathways **616** according to a preprogrammed sequence received in the instructions **606**. Thus, the control circuitry **630** can also control the species of nucleotide **614** that is added during any cycle of synthesis.

[0118] If the redox-pH control mechanism **622** is a microelectrode array **624**, the control circuitry **630** may be able to set the voltage independently at any (or all) of the electrodes in the microelectrode array **624** in any arbitrary pattern. The microelectrode array **624** may be any of the microelectrode arrays described in this disclosure. In an implementation, the microelectrode array **624** may be used to change the oxidation state of divalent metal cations complexed with His-tags attaching blocking groups to polymerases. For example, changes in electrode potential on the microelectrode array **624** may reduce Cu.sup.2+ to Cu.sup.+ . This will cause the blocking groups to be released and the polymerase becomes active. In an implementation, the microelectrode array **624** may be used the pH to an optimum pH for the polymerase thereby creating active polymerase. Additionally, a change in pH triggered by activation of the microelectrode array **624** may be used to release enzyme inhibitors from encapsulation.

[0119] If the redox-pH control mechanism **622** is a targeted fluid deposition instrument **626**, the control circuitry **630** may control the location of a print nozzle and the type of reagent that is dispensed onto the surface of the array **100**. Thus, the control circuitry **630** may cause an acid, a base, or a redox reagent, etc. to be dispensed according to any arbitrary pattern across one or more selected locations on the array **100**. In this implementation, the array **100** does not need to contain electrodes and may be an electrochemically inert surface.

[0120] The targeted fluid deposition instrument **626** may be implemented as any type of equipment or device that can precisely apply small volumes of chemical reagents to specific locations on the surface of the array **100**. Examples include a chemical inkjet printing device or precision laboratory robotics. Chemical inkjet printing uses techniques similar to conventional printing to place nanoliter volumes of reagents at specified locations on a two-dimensional surface. Techniques for using inkjet printing to precisely deliver chemical reagents to selected locations on a surface of an array are well-known to those of ordinary skill in the art. See Paul Calvert, *Inkjet Printing for Materials and Devices*, 13(10) Chem. Mater. 3299 (2001).

[0121] Any type of chemical inkjet printing may be adapted for use with this disclosure. Inkjet printing can be divided into two categories: (1) drop-on-demand (DoD) or impulse inkjet, where droplets are generated when required; and (2) continuous inkjet, in which droplets are deflected from a continuous stream to a substrate when needed. Inkjet printing can be further subdivided according to the specific means of generating droplets, such as piezoelectric, thermal, and electrostatic. Droplet size involves, typically, volumes ranging from 1.5 pL to 5 nL at a rate of 0-25 kHz for drop-on-demand printers (and up to 1 MHz for continuous printheads).

[0122] Electrohydrodynamic jet printing (EHJP) is another printing technology that may be used. EHJP is a high-resolution printing technology where the printed liquid is driven by an electric field. Exposure to an electric field causes mobile ions in a polarizable liquid to accumulate at the liquid surface. Deposited droplets can be as small as 240 nm with spatial accuracy in the hundreds of nm, which is typically an order of magnitude smaller than other inkjet printing technologies. Such small droplet sizes dispense less material with more spatial control, which allows for more selectivity in controlling polymerase activity.

[0123] In an implementation, the targeted fluid deposition instrument **626** may deliver a redox agent to the selected location on the array **100**. The redox reagent may be a reducing reagent or oxidizing reagent that changes the oxidation state of divalent metal cations complexed to a His-tag sequence. For example, the redox reagent may be a reducing agent such as a salt of ascorbic acid (e.g., sodium ascorbate), a salt of citric acid, a salt of sodium hypophosphate, a salt of hydrazine, or similar. Changing the oxidation state of the divalent metal cations to another oxidation state causes the blocking groups to be released from the His-tag sequence and this activates the polymerase.

[0124] The targeted fluid deposition instrument **626** may deliver an acid or a base that changes the

pH to an optimum pH for the polymerase thereby creating active polymerase **108** at the selected location. Addition of acid or base by the targeted fluid deposition instrument **626** may be used to create active polymerase **108** by changing the local pH to an optimum pH range for the enzyme. If the reaction reagent solution **612** is buffered to an unsuitable pH for the polymerase the polymerase may be denatured and inactive. Raising or lowering the pH to within the optimum pH range allows the polymerase to return to a native conformation and become active. Nucleotide polymerization is then possible at those locations where the targeted fluid deposition instrument **626** has added the acid or base. The buffering capacity of the reaction reagent solution **612** controls the range over which the added acid or base affects pH. As the buffering capacity increases, the area of effect for added acid or base decreases and spatial control becomes more precise.

[0125] In an implementation, the acid or base delivered by the targeted fluid deposition instrument **626** may release an enzyme inhibitor from encapsulation. Encapsulation systems for pH-dependent release of drugs or other small molecules are known to those of ordinary skill in the art and discussed above. Release of the enzyme inhibitor from encapsulation will cause the polymerase at the selected location to become inactive polymerase **106**.

[0126] If the redox-pH control mechanism **622** is a light source **628**, the control circuitry **630** may turn the light source **628** on and off and control where light from the light source **628** contacts the array **100**. Light from the light source **628** may be directed or focused on to the surface of the array **100** by optoelectronics such as a photomask or DMD. One example of a DMD that directs light onto an array surface is provided in Howon Lee et al. supra. The light source **628** generates light of a specific wavelength or range of wavelengths. Light from the light source **628** may be used to excite a photo-sensitive molecule such as a photoredox catalyst, a photoacid, or a photobase. In this implementation, the array **100** does not need to contain electrodes and may be an electrochemically inert surface.

[0127] In an implementation, light from the light source **628** is used to change the oxidation state of a divalent metal cation complexed with a His-tag. Light from the light source **628** can excite a photoredox catalyst that performs a photoredox reaction with the metal cation or an intermediary. The light may be of any spectrum that is capable of initiating a photochemical reaction that does not damage the polynucleotides or the template-independent polymerase. In an implementation, the light source **628** generates visible light. One suitable class of photoredox catalyst is the metal polypyridyl complexes of which iridium polypyridyl complexes are one example. These types of photoredox catalysts can perform single electron transfers (SET) as part of a series of reactions that reduce Co(III) to Co(II). See Megan H. Shaw et al., *Photoredox Catalysis in Organic Chemistry*, 81 J. Org. Chem. 5898 (2016). There are also suitable organic photoredox catalysts.

[0128] In an implementation, light from the light source **628** activates a photoacid or photobase and changes the pH at the selected location. The change in pH may generate active polymerase **108** by moving the pH at the selected location from an unsuitable pH for the enzyme to an optimum pH. Additionally, if polymerase activity is controlled by encapsulated enzyme inhibitors, the change in pH may be used to trigger location-specific release of the enzyme inhibitors from pH-dependent encapsulation.

Illustrative Embodiments

[0129] The following clauses described multiple possible embodiments for implementing the features described in this disclosure. The various embodiments described herein are not limiting nor is every feature from any given embodiment required to be present in another embodiment. Any two or more of the embodiments may be combined together unless context clearly indicates otherwise. As used in this document “or” means and/or. For example, “A or B” means A without B, B without A, or A and B. As used herein, “comprising” means including all listed features and potentially including addition of other features that are not listed. “Consisting essentially of” means including the listed features and potentially including additional features that do not materially affect the basic and novel characteristics of the listed features. “Consisting of” means only the

listed features to the exclusion of any feature not listed.

[0130] Clause 1. A method for enzymatic synthesis of polynucleotides comprising: (a) incubating an array having a plurality of initiators attached thereto with a reaction reagent solution comprising template-independent polymerase and with a selected species of nucleotide; and (b) altering redox-pH conditions at a selected location on the array such that activity of the template-independent polymerase is changed, without changing availability of a metal cofactor complexed with the template-independent polymerase.

[0131] Clause 2. The method of clause 1, wherein: the template-independent polymerase includes a His-tag sequence, the reaction reagent solution comprises blocking groups covalently modified with ligands and divalent metal cations that complex with the His-tag sequence and with the ligand, and wherein altering the redox-pH conditions comprises altering redox conditions such that an oxidation state of the divalent metal cation changes thereby releasing the blocking groups from complexes with the template-independent polymerase and increasing activity of the template-independent polymerase at the selected location.

[0132] Clause 3. The method of clause 2, wherein the divalent metal cations are selected from the group comprising nickel, cobalt, copper, or zinc.

[0133] Clause 4. The method of clause 2, wherein the divalent metal cation comprises Cu.sup.2+.

[0134] Clause 5. The method of any of clauses 2-4, wherein the blocking groups comprise proteins or polymers.

[0135] Clause 6. The method of clause 5, wherein the blocking groups have molecular weights that are between about 0.5-10× the molecular weight of the template-independent polymerase.

[0136] Clause 7. The method of clause 1, wherein the reaction reagent solution has an unsuitable pH for the template-independent polymerase, and wherein altering the redox-pH conditions comprises altering pH conditions to an optimum pH for the template-independent polymerase thereby increasing activity of the template-independent polymerase at the selected location.

[0137] Clause 8. The method of clause 1, wherein the reaction reagent solution comprises encapsulated inhibitors, and wherein altering the redox-pH conditions comprise altering pH conditions such that inhibitors are released from encapsulation at the selected location thereby decreasing activity of the template-independent polymerase at the selected location.

[0138] Clause 9. The method of any of clauses 1-8, wherein the array comprises a microelectrode array having a plurality of individually addressable electrodes and altering the redox-pH conditions at the selected location on the array comprises activating at least one of the individually addressable electrodes at the selected location.

[0139] Clause 10. The method of any of clauses 1-8, wherein the reaction reagent solution further comprises photoredox catalysts and altering the redox-pH conditions at the selected location on the array comprises exposing the selected location on the array to a wavelength of light that excites the photoredox catalyst.

[0140] Clause 11. The method of any of clauses 1-8, wherein altering the redox-pH conditions at the selected location on the array comprises delivering a chemical redox reagent, an acid, or a base, to the selected location with targeted fluid deposition instrument.

[0141] Clause 12. The method of any of clauses 1-11, wherein the selected species of nucleotide comprises unmodified nucleotides.

[0142] Clause 13. The method of any of clauses 1-12, further comprising: (c) contacting the array with a wash solution that removes the reaction reagent solution.

[0143] Clause 14. The method of clause 13, further comprising iteratively repeating steps (a), (b), and (c) such that both the selected location and the selected species of nucleotide change at least once between iterations.

[0144] Clause 15. The method of any of clauses 1-14, wherein template-independent polymerase is TdT.

[0145] Clause 16. The method of any of clauses 1-15, wherein plurality of initiators comprises

initiators having a length of between about 3-30 nucleotides.

[0146] Clause 17. The method of any of clauses 1-16, wherein the selected species of nucleotide is one of deoxyadenosine triphosphate (dATP), deoxycytidine triphosphate (dCTP), deoxyguanosine triphosphate (dGTP), deoxythymidine triphosphate (dTTP), adenosine triphosphate (ATP), cytidine triphosphate (CTP), guanosine triphosphate (GTP), or uridine triphosphate (UTP).

[0147] Clause 18. A method for enzymatic synthesis of a polynucleotide comprising: (a) incubating an array having a plurality of initiators attached thereto with a reaction reagent solution comprising inactive template-independent polymerase comprising a His-tag sequence complexed to a blocking group and with a selected species of nucleotide; and (b) altering redox conditions at a selected location on the array such that an oxidation state of a divalent metal cation complexed with the His-tag sequence changes thereby releasing the blocking group converting the inactive template-independent polymerase into active template-independent polymerase.

[0148] Clause 19. The method of clause 18, wherein the divalent metal cation comprises Cu.sup.2+.

[0149] Clause 20. The method of any of clauses 18-19, wherein the array comprises a microelectrode array having a plurality of individually addressable electrodes and altering the redox conditions at the selected location on the array comprises activating at least one of the individually addressable electrodes at the selected location.

[0150] Clause 21. The method of any of clauses 18-19, wherein the reaction reagent solution further comprises photoredox catalysts and altering the redox conditions at the selected location on the array comprises exposing the selected location on the array to a wavelength of light that excites the photoredox catalysts.

[0151] Clause 22. The method of any of clauses 18-19, wherein altering the redox conditions at the selected location on the array comprises delivering a chemical redox reagent to the selected location with a targeted fluid deposition instrument.

[0152] Clause 23. The method of any of clauses 18-22, further comprising: (c) contacting the array with a wash solution that removes the reaction reagent solution; and iteratively repeating steps (a), (b), and (c) such that both the selected location and the selected species of nucleotide change at least once between iterations.

[0153] Clause 24. A device for de novo synthesis of polynucleotides, the device comprising: an array having a plurality of initiators attached thereto; a first fluid delivery pathway configured to contact the array with a reaction reagent solution comprising a template-independent polymerase including a His-tag sequence complexed to a blocking group covalently modified with a ligand; a second fluid delivery pathway configured to contact the array with a selected species of nucleotide; and control circuitry configured to alter redox conditions at a selected location on the array changing an oxidation state of divalent metal cations complexed to the His-tag sequence thereby releasing the blocking group from the template-independent polymerase and to selectively open the first fluid delivery pathway and the second fluid delivery pathway.

[0154] Clause 25. The device of clause 24, wherein the divalent metal cations comprise Cu.sup.2+.

[0155] Clause 26. The device of any of clauses 24-25, further comprising a reaction chamber configured to maintain the reaction reagent solution in contact with the array.

[0156] Clause 27. The device of any of clauses 24-26, further comprising: a third fluid delivery pathway configured to contact the array with a second selected species of nucleotide; and a computing device communicatively coupled to the control circuitry and configured to sequentially activate a redox-pH control mechanism and sequentially open the second fluid delivery pathway and the third fluid delivery pathway according to a preprogrammed sequence.

[0157] Clause 28. The device of any of clauses 24-27, wherein the array comprises a microelectrode array having a plurality of individually addressable electrodes and the control circuitry is configured to alter redox conditions by activating at least one of the individually addressable electrodes at the selected location.

[0158] Clause 29. The device of any of clauses 24-27, wherein the reaction reagent solution further comprises photoredox catalysts and further comprising a light source configured to direct light to the photoredox catalysts at the selected location on the array.

[0159] Clause 30. The device of any of clauses 24-27, further comprising a targeted fluid deposition instrument configured to deliver a redox reagent to the selected location on the array.

CONCLUSION

[0160] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts are disclosed as example forms of implementing the claims.

[0161] The terms “a,” “an,” “the” and similar referents used in the context of describing the invention are to be construed to cover both the singular and the plural unless otherwise indicated herein or clearly contradicted by context. The terms “based on,” “based upon,” and similar referents are to be construed as meaning “based at least in part” which includes being “based in part” and “based in whole,” unless otherwise indicated or clearly contradicted by context. The terms “portion,” “part,” or similar referents are to be construed as meaning at least a portion or part of the whole including up to the entire noun referenced. As used herein, “approximately” or “about” or similar referents denote a range of $\pm 10\%$ of the stated value.

[0162] For ease of understanding, the processes discussed in this disclosure are delineated as separate operations represented as independent blocks. However, these separately delineated operations should not be construed as necessarily order dependent in their performance. The order in which the processes are described is not intended to be construed as a limitation, and unless otherwise contradicted by context any number of the described process blocks may be combined in any order to implement the process or an alternate process. Moreover, it is also possible that one or more of the provided operations is modified or omitted.

[0163] Certain embodiments are described herein, including the best mode known to the inventors for carrying out the invention. Of course, variations on these described embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. Skilled artisans will know how to employ such variations as appropriate, and the embodiments disclosed herein may be practiced otherwise than specifically described. Accordingly, all modifications and equivalents of the subject matter recited in the claims appended hereto are included within the scope of this disclosure. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

[0164] Furthermore, references have been made to publications, patents, and/or patent applications throughout this specification. Each of the cited references is individually incorporated herein by reference for its particular cited teachings as well as for all that it discloses.

Claims

1. A device for de novo synthesis of polynucleotides, the device comprising: an array having a plurality of initiators attached thereto; a first fluid delivery pathway configured to contact the array with a reaction reagent solution comprising a template-independent polymerase including a His-tag sequence complexed to divalent metal cations associated with a ligand attached to a blocking group; a second fluid delivery pathway configured to contact the array with a selected species of nucleotide; and control circuitry configured to (i) alter redox conditions at a selected location on the array changing an oxidation state of the divalent metal cations complexed to the His-tag sequence without changing availability of a metal cofactor complexed with the template-independent polymerase thereby releasing the blocking group from the template-independent polymerase and (ii) selectively open the first fluid delivery pathway and the second fluid delivery

pathway.

2. The device of claim 1, wherein the first fluid delivery pathway and the second fluid delivery pathway are both independently implemented by one or more of tubes and pumps, microfluidics, and laboratory robotics.
 3. The device of claim 1, wherein the selected species of nucleotide comprises unmodified nucleotides.
 4. The device of claim 1, wherein the metal cofactor comprises Co.sup.2+.
 5. The device of claim 1, wherein the divalent metal cations comprise Cu.sup.2+.
 6. The device of claim 1, further comprising a reaction chamber configured to maintain the reaction reagent solution in contact with the array.
 7. The device of claim 1, further comprising a third fluid delivery pathway configured to contact the array with a second selected species of nucleotide.
 8. The device of claim 7, further comprising a computing device communicatively coupled to the control circuitry and configured to sequentially activate a redox-pH control mechanism and sequentially open the second fluid delivery pathway and the third fluid delivery pathway according to a preprogrammed sequence.
 9. The device of claim 8, wherein the computing device further comprises a polynucleotide synthesizer control module configured to receive input specifying a target nucleotide sequence and generate a preprogrammed sequence of operations comprising sequential activation of the redox-pH control mechanism and selective opening of the first fluid delivery pathway and the second fluid delivery pathway to synthesize the target nucleotide sequence on the array.
 10. The device of claim 8, wherein the redox-pH control mechanism is one of a microelectrode array, a targeted fluid deposition instrument, or a light source.
 11. The device of claim 1, wherein the array comprises a microelectrode array having a plurality of individually addressable electrodes and the control circuitry is configured to alter the redox conditions by activating at least one of the individually addressable electrodes at the selected location.
 12. The device of claim 11, wherein the microelectrode array has a microelectrode density of at least 1024 microelectrodes/cm.sup.2 or at least 12,544 microelectrodes/cm.sup.2.
 13. The device of claim 11, wherein the microelectrode array comprises complementary metal-oxide-semiconductor (CMOS) electrodes.
 14. The device of claim 1, wherein the reaction reagent solution further comprises photoredox catalysts and further comprising a light source configured to direct light to the photoredox catalysts at the selected location on the array.
 15. The device of claim 14, wherein the photoredox catalysts comprise metal polypyridyl complexes.
 16. The device of claim 14, further comprising a photomask or digital micromirror device (DMD) configured to direct the light source.
 17. The device of claim 1, further comprising a targeted fluid deposition instrument configured to deliver a redox reagent to the selected location on the array.
 18. The device of claim 17, wherein the targeted fluid deposition instrument comprises a chemical inkjet printing device.
 19. The device of claim 18, wherein the redox reagent comprises ascorbic acid, citric acid, sodium hypophosphate, or hydrazine.
 20. The device of claim 1, further comprising a fourth fluid delivery pathway configured to contact the array with a wash solution.
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